UNDERGRADUATE RESEARCH · HOW TO

How do you run a **cross-sectional study** on NCD risk factors?

What I learnt doing one in medical college, including what I got wrong.

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STEP 1 · THE DESIGN

A cross-sectional study is a **snapshot.**

You look at a population at one point in time. You measure risk factors and conditions together.



It tells you how common something is. It cannot tell you what caused what.

STEP 2 · THE TOOL

Do not invent your questions. **Use a framework.**

1 **Questionnaire**
Tobacco, alcohol, diet, physical activity

2 **Physical measurements**
Height, weight, waist, blood pressure

3 **Biochemical tests**
Blood glucose, cholesterol

We ran Steps 1 and 2.

SOURCE: WHO STEPS, the stepwise approach to NCD risk factor surveillance.

STEP 3 · THE PROTOCOL

Draft the **protocol** first.

1 **Question and objectives**
What exactly are you asking?

2 **Design, setting, population**
Who, where, and how many?

3 **Tools and variables**
What will you measure, and how?

4 **Analysis plan**
Which tables will the paper need?

5 **Ethics and consent**
Who approves it, and how do you take consent?

STEP 4 · APPROVALS

No data before **approval.**

Take the protocol to the Institutional Research Committee and the Ethics Committee. We had both before we collected anything.

Approval comes before the first participant, not after the first spreadsheet.

STEP 5 · THE ANALYSIS PLAN

Plan the analysis **before** the data.

Decide your variables, how you will code them, and what the empty tables look like. Draw the tables before you collect a single row.

I did not do this. That is the part of the story that cost me.

HOW MINE STARTED

It started on **the wards.**

In second year I went department by department, ward by ward. I tried to understand why the diet books were always empty, and filled up with renal diets and milk and eggs.

Dr. Asma Rahim and Dr. Ashik Refeek told me to narrow it down.

THE OPPORTUNITY

A health camp. **Why waste the data?**

The then College Union was running a health camp. A lot of tests would be done.

Why not assess the risk factors of all the students on campus? That is how we drafted the WHO STEPS 1 and 2 protocol.

WHAT WENT WRONG

We collected first. **We understood later.**

It took another one and a half years to get into the data, and learn to process it with SPSS and Excel.

That took a lot of my time in the final year, particularly the eighth semester.

THE MISTAKE

We spread **ourselves too thin.**

Too much data. Far too much data. Unsegmented, without the tools or the team to understand it, structure it properly or even publish it.

THE OUTCOME

An award, and **no paper.**

We presented a report at national forums and undergraduate conferences. The project earned me the KUHS UG Research Award.

#	PROJECT	OUTPUT	STATUS
1	**WHO STEPS 1 & 2 risk-factor survey (NCD)**	Presented at national forums and UG conferences	NOT PUBLISHED
2	**Nutritional assessment of cancer patients**	Indian J Med Paediatr Oncol, Mar 2026 (2nd author)	PUBLISHED
3	**Metastatic solitary fibrous tumour, parapharyngeal region**	Indian J Med Paediatr Oncol, Apr 2025 (6th author)	PUBLISHED
4	**Juvenile justice and recidivism (elective)**	Report and forums; not pushed for publication	NOT PUBLISHED
5	**Obesity-driven cardiac mortality, US adults 15 to 44**	JAHA, Oct 2026 (4th author)	PUBLISHED
6	**Retrosternal goitre case series**	Ongoing	IN PROGRESS

I was never able to publish it.

DO IT BETTER

If I did it again:

1 **One question**

Not every risk factor at once.

2 **Fewer variables**

Only what you will analyse.

3 **Analysis plan first**

Empty tables before data.

4 **A mentor who has published**

Before you collect, not after.

Next: R03. How to do research on real patients.